

34870 Lab E - Loudspeaker response in free field

Rooms 028/025 – Building 354

Purposes:

a) Illustrate how the frequency response and directivity of a loudspeaker unit depend on the shape of the baffle it is mounted in.

b) Measure the frequency responses of the units in the project loudspeaker system.

Equipment:

- Desktop PC with internal sound card,
- UMIK MiniDSP microphone & USB cable,
- Audio amplifier,
- 3" full range driver in a box,
- Circular and rectangular screens,
- Project loudspeaker system with vent plugs,
- Switchboard with three switches.

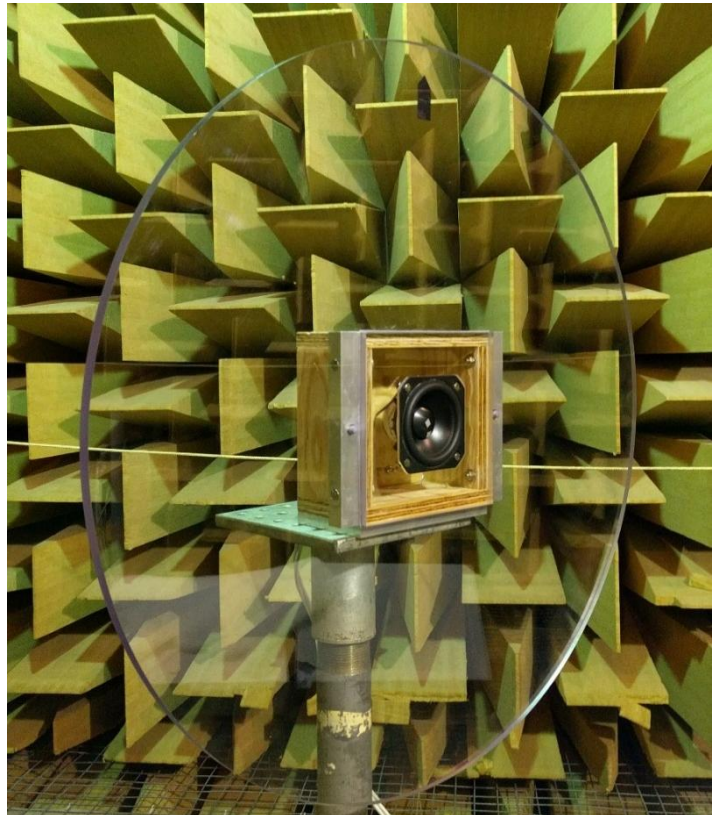


Figure 1 - Loudspeaker with the 3" driver (part 1) in its circular baffle

Warnings:

- **The absorbing wedges in the anechoic chamber are very fragile and expensive.** Pay extra attention not to touch and damage them, especially when you move equipment in and out of the room.
- Be careful not to drop screws, nuts or small objects through the net in the chamber when setting up the loudspeaker stand and screens. **They cannot be recovered.**
- **Loudspeaker:** The moving parts of the driver units are fragile, especially the tweeter which can be damaged even by a light touch. **Please do not touch the drivers.**
- Your loudspeaker system consists of two boxes: a woofer box and a mid/tweeter box. There are six systems named with letters A to F. The mid/tweeter box has a vent that must be kept closed; only the vents in the woofer box are used for low frequency design. You must use the same system for the rest of the course. Some groups may have to share systems: avoid booking a slot in parallel with another group using your system.

Part 1: Effect of the mounting shape

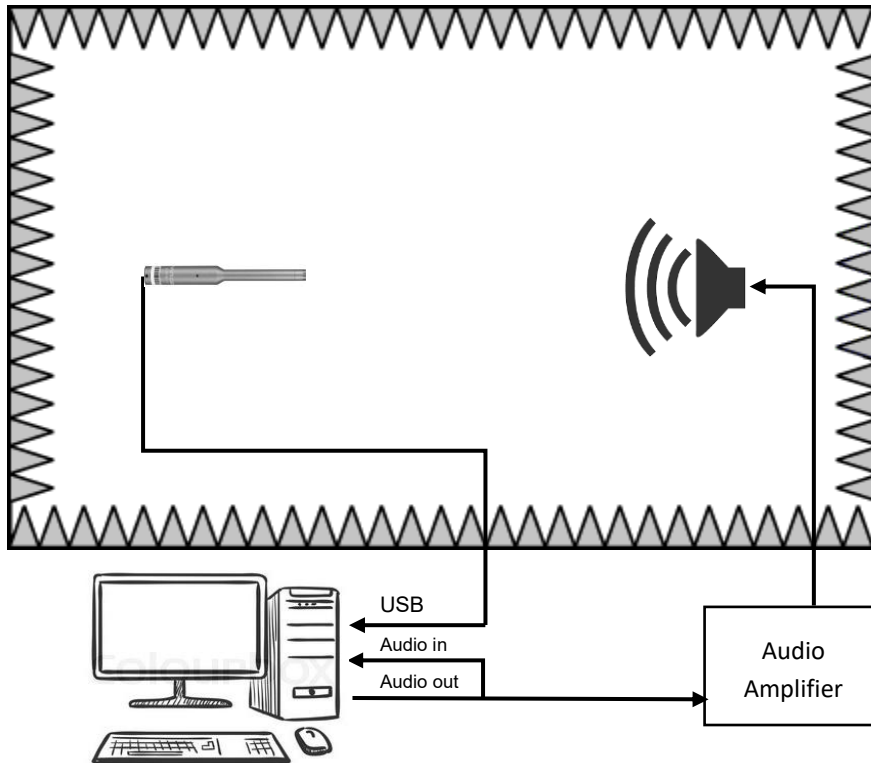


Figure 2: Sketch of the measurement setup

The mounting and environment around a loudspeaker can significantly affect its radiation pattern. Therefore, the measurement conditions are just as important as the driver itself. The ideal condition for measuring a loudspeaker driver is when mounted in an infinite baffle. This can be achieved in a semi-anechoic chamber, i.e. with one fully reflecting surface, but in a full anechoic room some approximation must be done. In this section it is shown how an asymmetric mounting on a finite size baffle helps minimizing the diffraction coming from the edges. Fig. 1 illustrates this part of the exercise.

Measurement procedure/routines

The exercise uses the internal sound card of the desktop PC in the control room. The “line-out” jack has been fed back to the “line-in” jack and sent to the audio amplifier input. The UMIK microphone should be properly placed in the anechoic chamber (see Fig. 2) and connected with the extended USB cable to a USB port in the desktop. Do this before opening Matlab so that the device is registered. You will need to run some trial measurements to find the proper volume levels of the amplifier. Adjust the volume control in the audio amplifier (starting very low) until the amplitude of the measured responses is well over noise. Time signals should be less than one. Use the plots generated by the measurement function for these verifications.

You will use the Matlab measurement function `meas_mag_spec2_SoundCard_LabE`. The necessary files are distributed through DTU Learn (Lab E Module) and are also available in the lab PC. When you have all the data, you may collect the results and delete the files from the computer. Matlab command:

```
[fn,specn,f,spec,ch]=
meas_mag_spec2_SoundCard_LabE(f1,f2,n_oct,fres,Nav,UmikSN,IncAngle,Nmeas)
```

The outputs are:

```
fn :      signal frequency vector
specn:    spectrum values for fn
f:        full frequency vector
spec:     full resolution spectrum
ch:       average timesignals
```

Use the following parameters. Keep the values for all Part 1.

```
f1=50 Hz           (low frequency limit)
f2=20000 Hz        (high frequency limit)
n_oct=24           (number of frequencies pr octave)
fres=1 Hz          (Spectrum frequency resolution)
Nav=16;            (Long averaging to suppress noise)
UmikSN;            (Umik serial number as a string. e.g. '708-0335')
IncAngle=0;        (Angle of incidence on the measurement microphone)
Nmeas=1;           (Number of consecutive measurements, 1-3)
```

Measure the effective diameter of the 3-inch driver. This means including half of the surround. Alternatively, a rule of thumb says that the piston radius in cm is roughly equal (numerically) to the advertised diameter in inches of the unit.

a) Frequency response – small box:

- Measure the loudspeaker in its small box in the frequency range 50 Hz -20 kHz.
- Measure at 0 and 30 degrees (horizontal). You can turn on its thread the metal bar over which the box is mounted to change the angle.
- Calculate the transfer function between the applied voltage and the measured sound pressure:

$$H_{pv} = \text{specn}(:,2) ./ \text{specn}(:,1)$$

b) Frequency response - circular screen:

- Mount the circular screen (black arrow should point upwards).
- Measure the screen radius.
- Measure at 0 and 30 degrees (horizontal).
- Calculate the pressure/voltage transfer functions.

c) Frequency response – IEC screen:

- Mount the rectangular screen with asymmetric speaker position (the orientation should place the loudspeaker in the upper part)
- Measure the screen dimensions
- Measure at 0, 15, 30 and 60 degrees (horizontal)
- Calculate the pressure/voltage transfer functions

Analysis part 1:

- Observe the differences in the frequency response for the three different mountings.
- For the circular screen, identify the relation between the screen dimensions and the pattern in the frequency response for on- and off-axis. Use the theory.
- For the asymmetric baffle only: Normalize the off-axis response with the on-axis measurement and compare with the directivity of a piston in a baffle.
- Calculate the frequency response for the circular baffle, using the `difffrac.m` Matlab function. Do this for both a point source and for an extended source with the radius you have measured earlier. In practice, you may set a very small source radius as a point source. Define many corner positions along the circular edge (follow the instructions in the function's help). It is also possible to calculate the IEC screen using `difffrac.m` by defining the corner sources along the edges of the rectangle.

Part 2: Project system anechoic measurement

a) Measure the effective diameter of each driver/unit as explained in Part 1.

b) Measure the frequency responses of the drivers in the box. The procedure using the sound card and UMIK microphone is described in the subsection c) below. These measurements of magnitude and phase are needed for a proper design of the crossover filters in the loudspeaker project.

- Unmount the small speaker cabinet used in part 1 and its support structure from the wooden platform (take care not to drop the nuts and bolts through the holes in the net: screw them on the spare holes).
- Place the speaker system on the platform.
- The position of the smaller box with respect to the woofer box affects the result. You may have to try several positions to get the best response. Check also if you want to keep the dust cover on. You should keep the same settings for the project.
- Position the microphone at the height where you want to be listening to the loudspeaker.
- Keep this microphone position: this allows for a correct measurement of the phase differences between the drivers.
- Measure the distance from the loudspeaker to the microphone. Make it a single rough estimation of the distance, not to a drive unit in particular. You may use it as a starting guess for the procedure in subsection d) below.
- Block the vent of the woofer. Remember that the anechoic chamber is not suitable for very low frequency, therefore the vent effect is not relevant here.
- Increase the frequency resolution to 1/48 octave ($n_{oct}=48$).
- Increase the averaging ($N_{av}=64$).
- The frequency range is set to 50 Hz - 20 kHz.
- Measure the frequency response of the three drivers at 0, 30 and 60 degrees horizontally. You can turn the wooden base to the desired angle.

c) Measurement procedure with the sound card:

The measurement with the UMIK microphone and the sound card introduces a random phase delay that can be different if the three units are measured separately. We need the relative phases between units to be accurate for the crossover design. We therefore run all three measurements in a single measurement sequence. A switchboard with three switches, one input and three outputs, is used. The three units are connected to the three outputs. Set $N_{meas}=3$ when you call the measurement function. The function will then perform three consecutive measurements separated by two intervals of 30 seconds warning sounds. Use the intervals to switch to the next unit in the switchboard. After the measurement, the variable `specn` contains all three results from the two channels. You may obtain the three responses by calculating:

```
H_1 = specn(:,2,1)./specn(:,1,1);
H_2 = specn(:,2,2)./specn(:,1,2);
H_3 = specn(:,2,3)./specn(:,1,3);
```

d) Procedure for obtaining a compensated plot of relative phases between the drivers at 0 degrees:

Besides the sought-for relative phases inherent to the units, the measured phases of the three responses are affected by two types of phase shifts:

- The distance between loudspeaker units and microphone introduces a phase shift that grows with frequency.
- The measurement with the UMIK microphone and sound card does not synchronize excitation and response and introduces unknown phase shifts. The measurement procedure described in subsection c), however, maintains the relative phases we are interested in.

The following procedure will find a compensation “distance” to correct the phase plot and make the relative phases more visible:

- 1- Start by running the `phaseunwrap.m` function in Matlab to unwrap the measured phase responses and introduce a compensating distance, the same for all units. You may just set $d=0$, or else use the rough distance estimate from subsection **b)**. Represent the compensated phases of the three responses.
- 2- If all phases grow with frequency, you need to try again with a shorter distance (even a negative value, if needed). Always use the same distance for all units.
- 3- If all phases decrease with frequency, you need to try with a longer distance. Again, use the same distance for all units.
- 4- After some iterations with progressively smaller distance variations, the units will present increasing and decreasing phases, or even almost constant phase for some unit. You are finished at this point. You have found an appropriate compensation distance, and your plot can show relative phases more clearly.

Remember that you can always add or subtract integer multiples of 2π radians to a phase response to make all phases move closer. This has no effect.

Evaluation of the exercise

The exercise work is made in groups of ideally three students. However, the evaluation is individual using a quiz in DTU Learn. The responses do not need to be the same for all students in the group. The quiz covers Lab D and Lab E. Submit the quiz as soon as possible and **no later than the deadline Monday 26th October**.